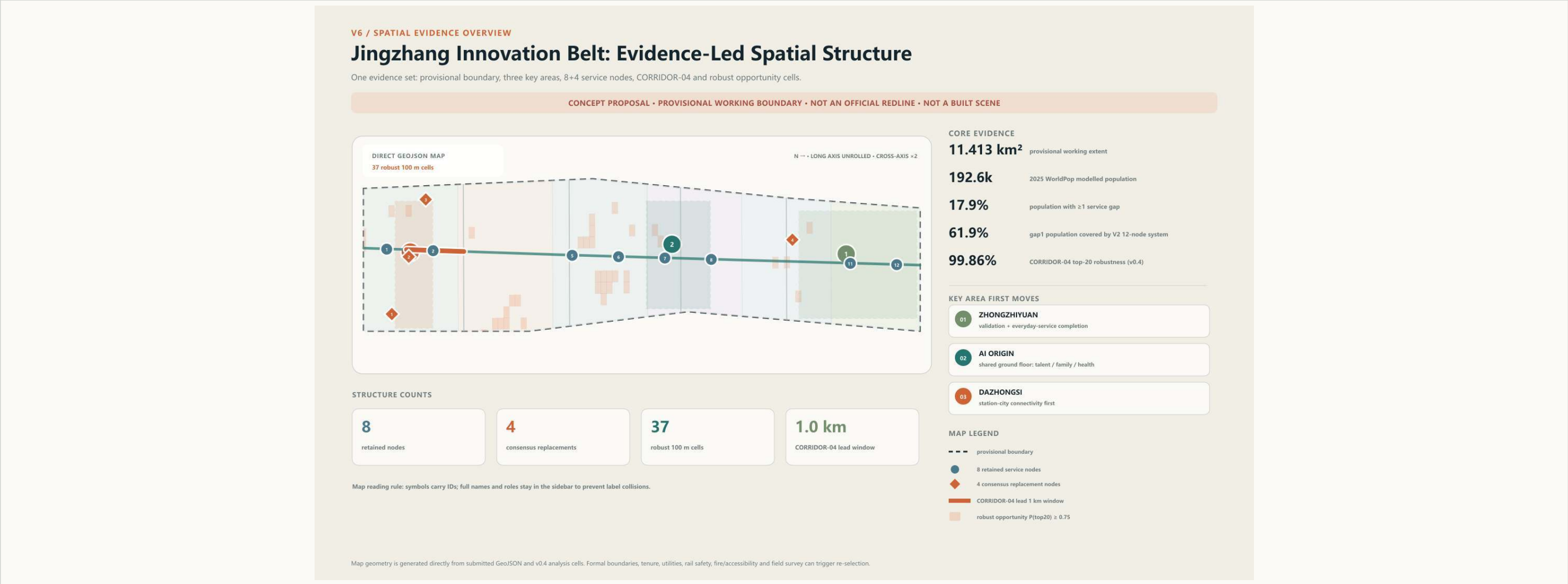


CENTENNIAL JINGZHANG AI INNOVATION BELT

FROM URBAN DIAGNOSIS TO AUDITABLE DESIGN



11.41 km²
provisional overall-design base

192.6k
WorldPop modelled population

17.9%
population with ≥1 service gap

61.9%
V2 consensus gap1 coverage

Existing Evidence and Overall Spatial Structure

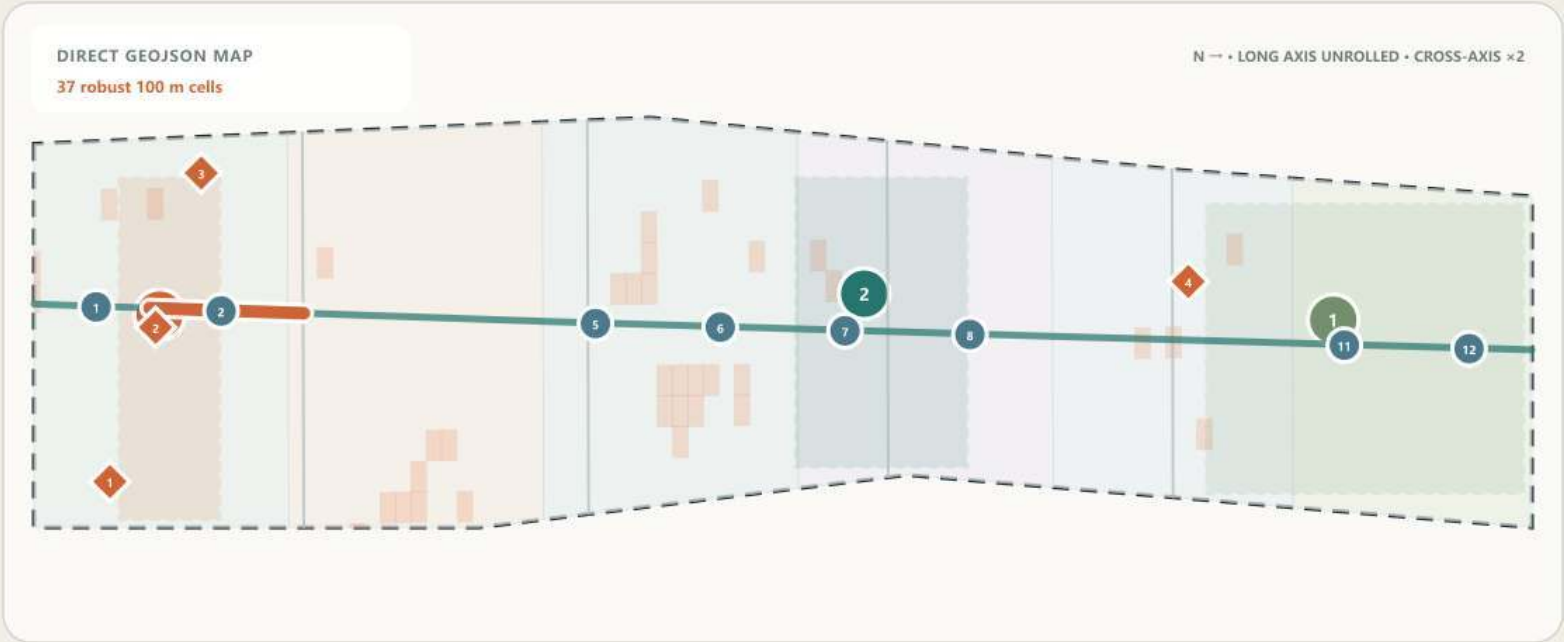
One evidence base constrains both diagnosis and concept design

V6 / SPATIAL EVIDENCE OVERVIEW

Jingzhang Innovation Belt: Evidence-Led Spatial Structure

One evidence set: provisional boundary, three key areas, 8+4 service nodes, CORRIDOR-04 and robust opportunity cells.

CONCEPT PROPOSAL • PROVISIONAL WORKING BOUNDARY • NOT AN OFFICIAL REDLINE • NOT A BUILT SCENE



CORE EVIDENCE

- 11.413 km² provisional working extent
- 192.6k 2025 WorldPop modelled population
- 17.9% population with ≥1 service gap
- 61.9% gap1 population covered by V2 12-node system
- 99.86% CORRIDOR-04 top-20 robustness (v0.4)

KEY AREA FIRST MOVES

- 01 ZHONGZHIYUAN validation + everyday-service completion
- 02 AI ORIGIN shared ground floor: talent / family / health
- 03 DAZHONGSI station-city connectivity first

STRUCTURE COUNTS

8

retained nodes

4

consensus replacements

37

robust 100 m cells

1.0 km

CORRIDOR-04 lead window

Map reading rule: symbols carry IDs; full names and roles stay in the sidebar to prevent label collisions.

MAP LEGEND

- provisional boundary
- 8 retained service nodes
- 4 consensus replacement nodes
- CORRIDOR-04 lead 1 km window
- robust opportunity P(top20) ≥ 0.75

WorldPop modelled population, OSM walk network, robust opportunity and promoted V2 geometry are shown together. Map geometry is generated directly from submitted GeoJSON and V2 analysis cells. Formal boundaries, tenure, utilities, rail safety, fire/accessibility and field survey can trigger re-selection.

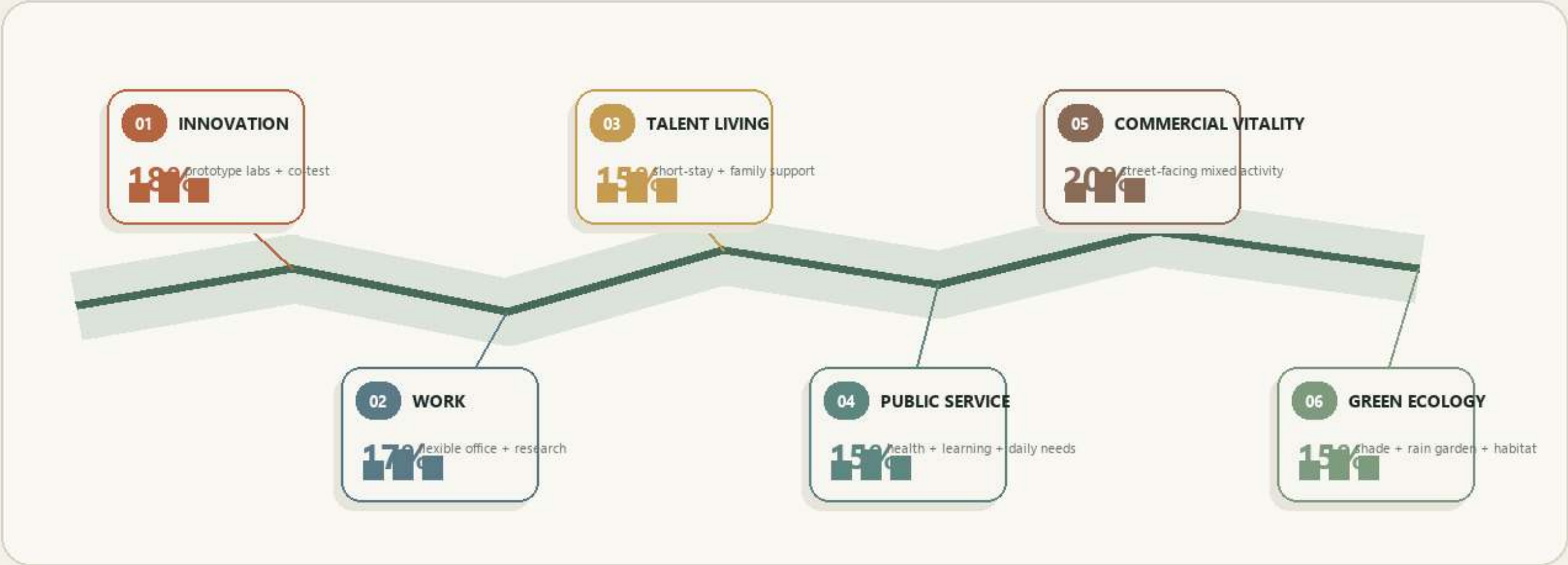
Six Mixed-use Functions Grow Along One Spine

Innovation, work, talent living, public services, commerce and green ecology interweave

02 / MIXED-USE SPINE

Six Mixed-Use Functions Grow Along One Spine

Innovation, work, talent living, public services, commerce and green ecology interweave within one linear corridor - not six isolated zoning blocks.



PROGRAM MIX



18% innovation 17% work 15% talent living 15% public service 20% commercial vitality 15% green ecology

Conceptual program mix for design comparison; subject to statutory land-use controls, tenure, building surveys and implementation-stage verification.

Concept program mix: 18% / 17% / 15% / 15% / 20% / 15%; for design comparison and subject to statutory land-use, tenure, building surveys and implementation-stage verification.

The Three Key Areas Require Different Diagnoses

Population-weighted mobility, health and service gaps define the first move

03 / THREE URBAN CASES

THREE KEY AREAS, THREE DIFFERENT FIRST MOVES

Population-weighted mobility, health and service gaps define three distinct first-stage spatial prototypes.

ZHONGZHIYUAN

01

VALIDATION + EVERYDAY-SERVICE COMPLETION



FIRST-STAGE PROJECT

The ~1.5 ha test field is a research hypothesis and must be checked together with real everyday-service deficits.

Concept visualization / not factual evidence

AI ORIGIN COMMUNITY

02

SHARED GROUND FLOOR: TALENT / FAMILY / HEALTH



FIRST-STAGE PROJECT

Mobility is comparatively mature; prioritize learning, talent/family support and health referral.

Concept visualization / not factual evidence

DAZHONGSI

03

STATION-CITY CONNECTIVITY FIRST



FIRST-STAGE PROJECT

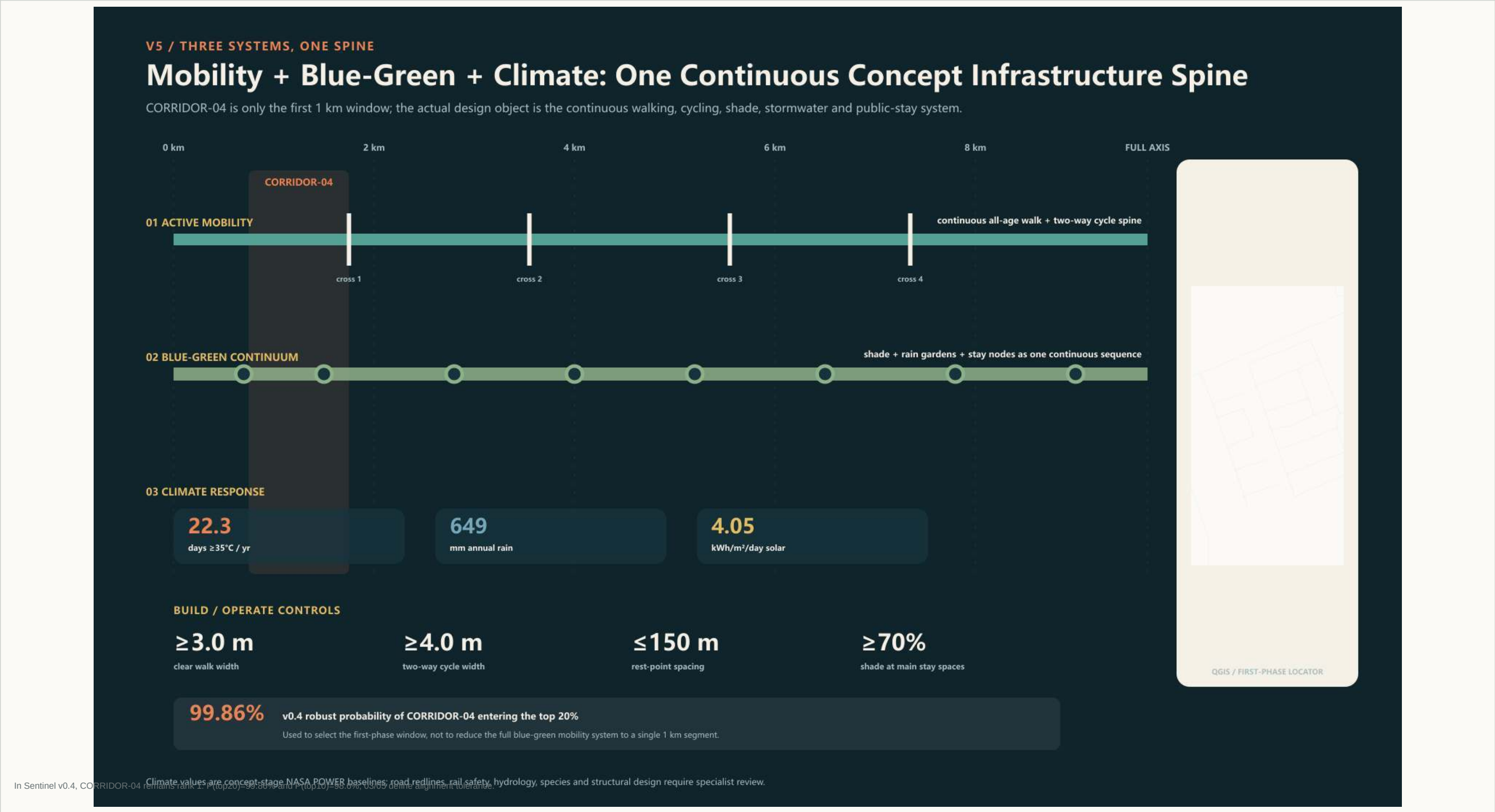
Low rail access, high service gaps and the largest detour proxy put walking/cycling connectivity first.

Concept visualization / not factual evidence

Zhongzhiyuan completes services, AI Origin focuses talent/family support, and Dazhongsi repairs station-city connectivity first.

Why the One-kilometre Pilot Starts on the Dazhongsi Side

36 sliding windows + 5,000 weight samples



Evidence Contract: From Conditions to Design, Impact and Acceptance

Model/remote-sensing evidence, spatial response, expected impact and verification conditions share one chain

V6 / METRIC CONTRACT

Four Evidence Classes: Baseline, Design, Scenario, Delivery

Each number keeps its evidence type and verification condition; scenario effects are not built outcomes.

01 BASELINE / MODEL

192.6k
2025 WorldPop modelled population

17.9%
population with ≥1 service gap

26.23%
Sentinel-2 vegetation spectral proxy

02 DESIGN RESPONSE

12
service-node modules

320 m²
prototype area per node

1.0 km
CORRIDOR-04 first-phase window

03 SCENARIO EFFECT

39.4→61.9%
gap1 coverage: V1 → V2

51.5%
coverage of population with ≥2 gaps

99.86%
CORRIDOR-04 top-20 robustness

04 DELIVERY CONTROL

≥3 / ≥4 m
walk / two-way cycle clear width

≤150 m
effective rest-point spacing

≥70%
shade target at main stay spaces

100%
human-service availability target

MEASURE → AUDIT → TRIGGER → ACCEPT

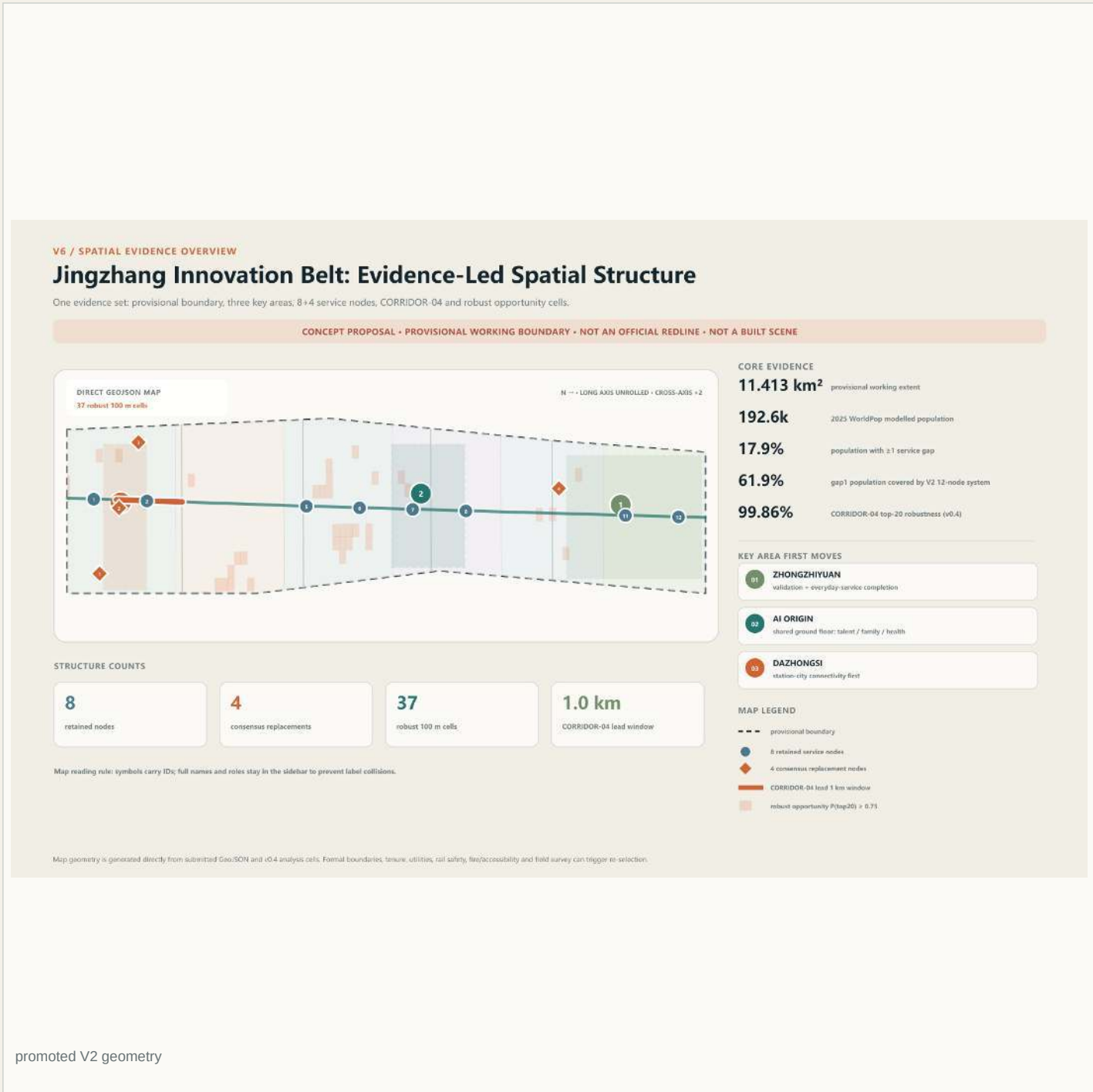


RE-CHECK TRIGGERS
formal redlines / tenure / utilities / rail safety / fire + accessibility / official facility data / field survey

With the same 12-node count, modelled total population coverage rises from 35.0% to 57.0% and gap1 from 39.4% to 61.9%; CORRIDOR-04 has P(top20)=99.86%. These are model/design-stage evidence, not built outcomes.

Overall Structure: Spatial Evidence and Mixed-use Programming Work Together

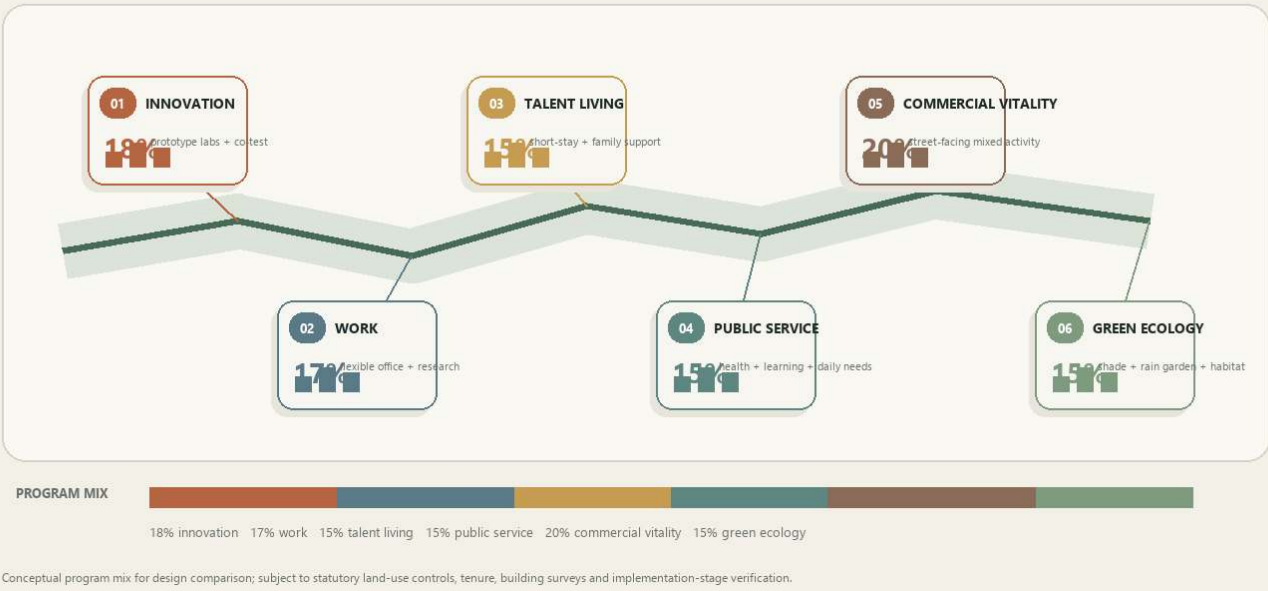
Concept design remains subject to formal redlines, tenure, utilities and field checks



02 / MIXED-USE SPINE

Six Mixed-Use Functions Grow Along One Spine

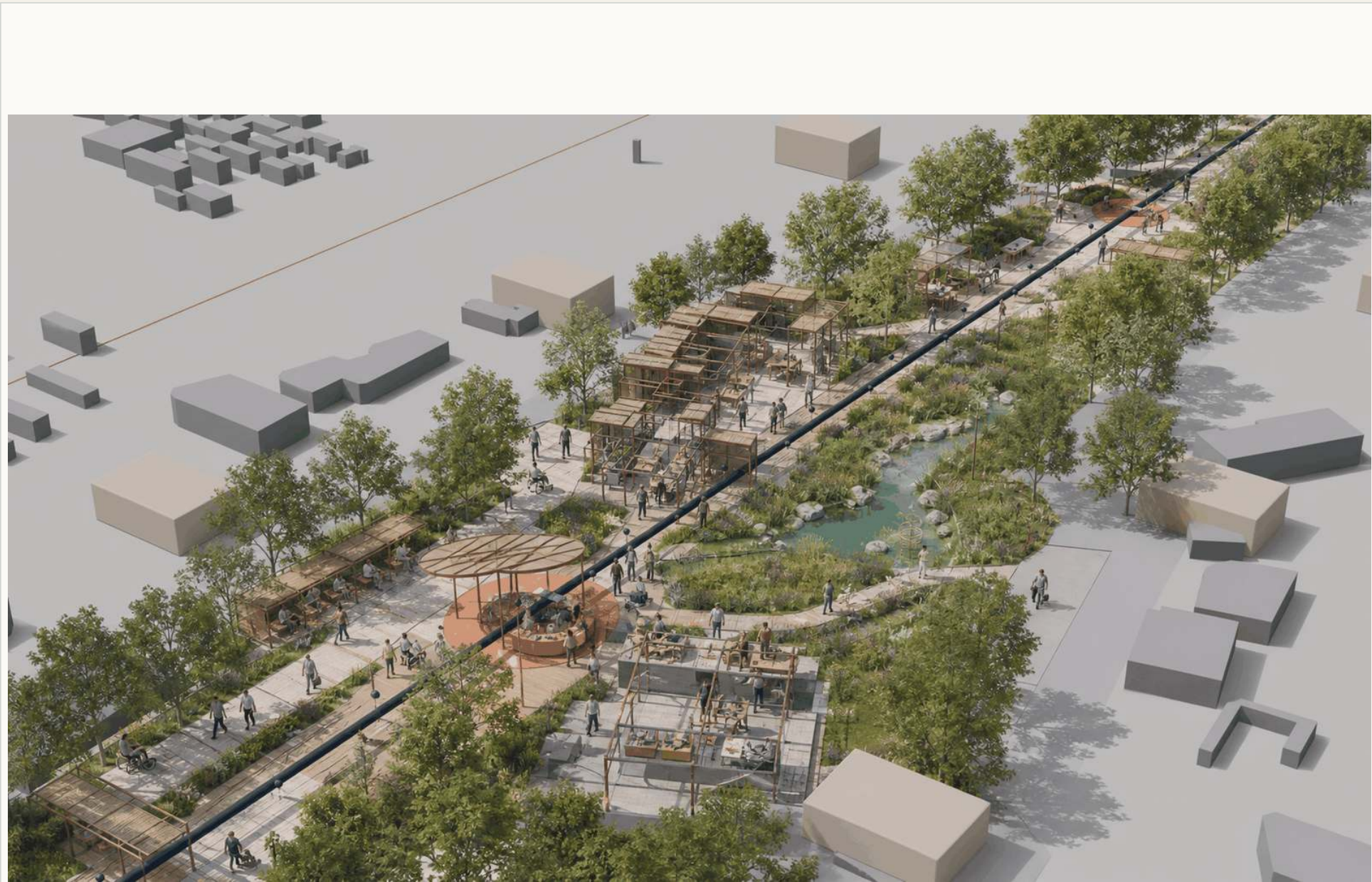
Innovation, work, talent living, public services, commerce and green ecology interweave within one linear corridor - not six isolated zoning blocks.



six mixed-use functions

Zhongzhiyuan | Couple Validation with Everyday-service Completion

The ~1.5 ha test field remains a research hypothesis but must be re-sited and checked against real service deficits.



Concept visualization / not factual evidence

22.6k
modelled population

53.0%
rail within 10 min

79.5%
health within 15 min

36.8%
gap in ≥1 service

1.94×
pedestrian detour proxy

AI Origin | Shift the Shared Ground Floor toward Talent and Life Support

Mobility is relatively strong; the first stage should emphasize talent, family, health referral, learning/collaboration and open innovation.



Concept visualization / not factual evidence

14.9k
modelled population

86.7%
rail within 10 min

63.5%
health within 15 min

36.5%
gap in ≥ 1 service

1.71x
pedestrian detour proxy

Dazhongsi | Repair Station-city Connectivity before Expanding the Public Hall

Low 10-minute rail coverage, high health deficit and the largest detour proxy place connectivity ahead of the 2.4 ha public-hall expansion.



Concept visualization / not factual evidence

18.0k
modelled population

25.3%
rail within 10 min

49.6%
health within 15 min

53.6%
gap in ≥ 1 service

2.78x
pedestrian detour proxy

320 m² Service Node: Small, Demountable and Staffed

Total concept footprint is 3,840 m²; module size is not an approved construction area

10 / SERVICE NODE GUIDE

Public-service Node Delivery and Operations Guide

A 320 m² prototype defines site, program, staffing, utilities, acceptance and exit requirements



NODE DELIVERY + OPERATIONS

250–400 m² program range | 320 m² prototype

01	FOOTPRINT	320 m²
	20 m × 16 m prototype with demountable components	
	ACCEPT site and built-area check	
02	HUMAN SERVICE	24–36 m²
	counter, waiting, storage and staffed position	
	ACCEPT hours and staffing roster	
03	MICROCLIMATE	≥70%
	continuous shade across principal stay areas	
	ACCEPT summer solar study + field audit	
04	RESPONSE	60 s / 24 h
	human acknowledgement and routine fault logging	
	ACCEPT monthly response + work-order log	

HANDOVER + EXIT MANAGEMENT

- joint utilities, fire, access and site-permit review
- facility list, duty register and monthly inspection record
- remove equipment if operations lapse; retain basic civic amenities

The four new sites still require official-facility overlap, tenure, utilities, fire, accessibility and field checks. generated-service-node-axonometric.jpg

node delivery integrates utilities, fire review, accessibility and operating maintenance

Pilot Section: Translate Accessibility into Design Controls

clear walk ≥3.0 m; two-way cycle ≥4.0 m; effective rest spacing ≤150 m

09 / TYPICAL SECTION

Typical-section Controls for the Jingzhang Pilot

A 12-m typical section addresses walking gaps, mixed traffic, limited rest and stormwater drainage



SECTION CONTROLS + ACCEPTANCE

12.0 m total | four continuous functional bands

01	WALKING	≥3.0 m
	continuous clear width, tactile and kerb-ramp links	
	ACCEPT field width survey	
02	STORMWATER	≥2.5 m
	complete inflow, overflow and drainage route	
	ACCEPT hydraulic check + flow test	
03	CYCLING	≥4.0 m
	two-way operation with speed control at crossings	
	ACCEPT marking width + conflict review	
04	SERVICE EDGE	≥2.5 m
	seating, shade and staffed wayfinding	
	ACCEPT facility list + access audit	

DELIVERY INTERFACES

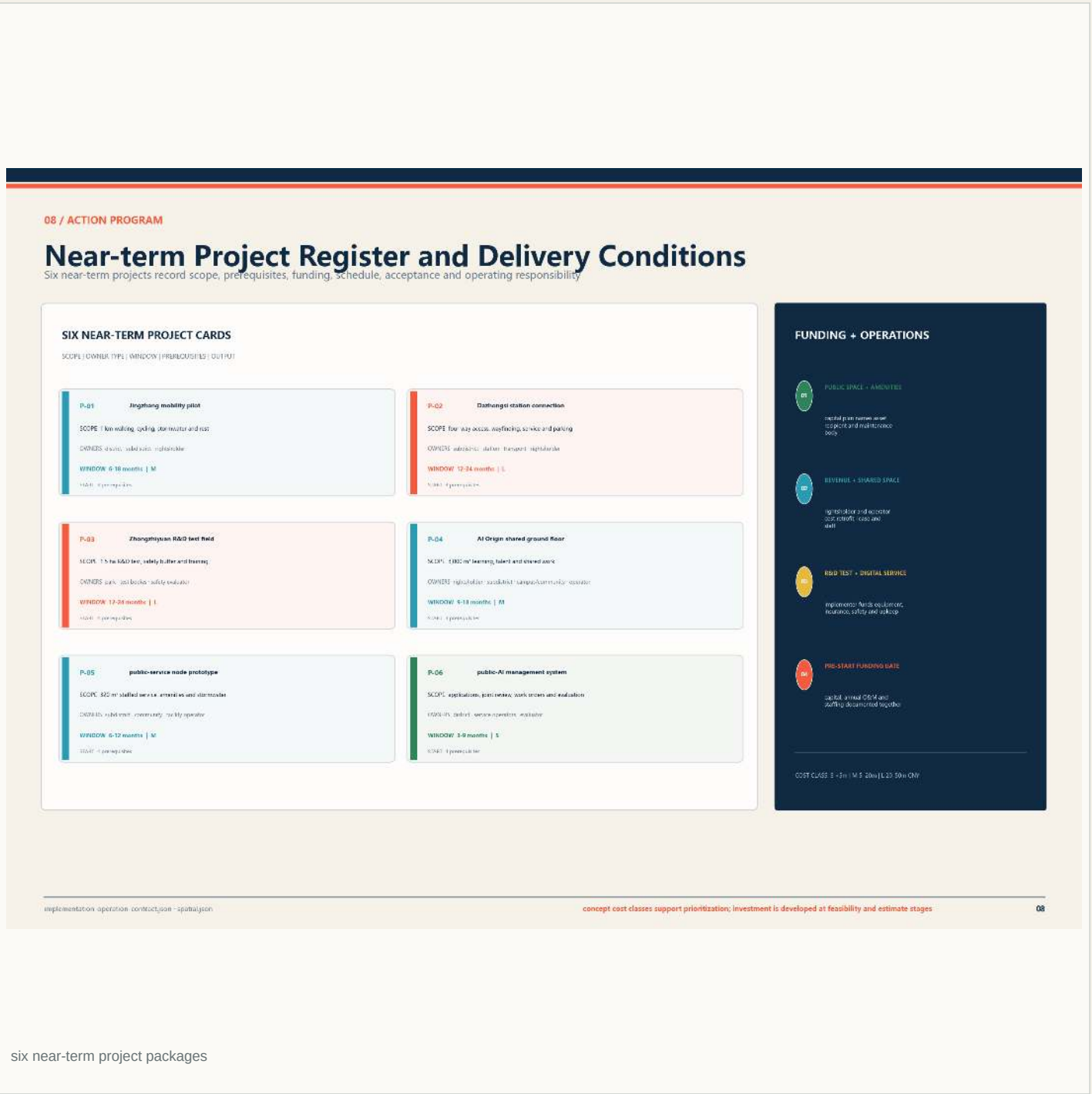
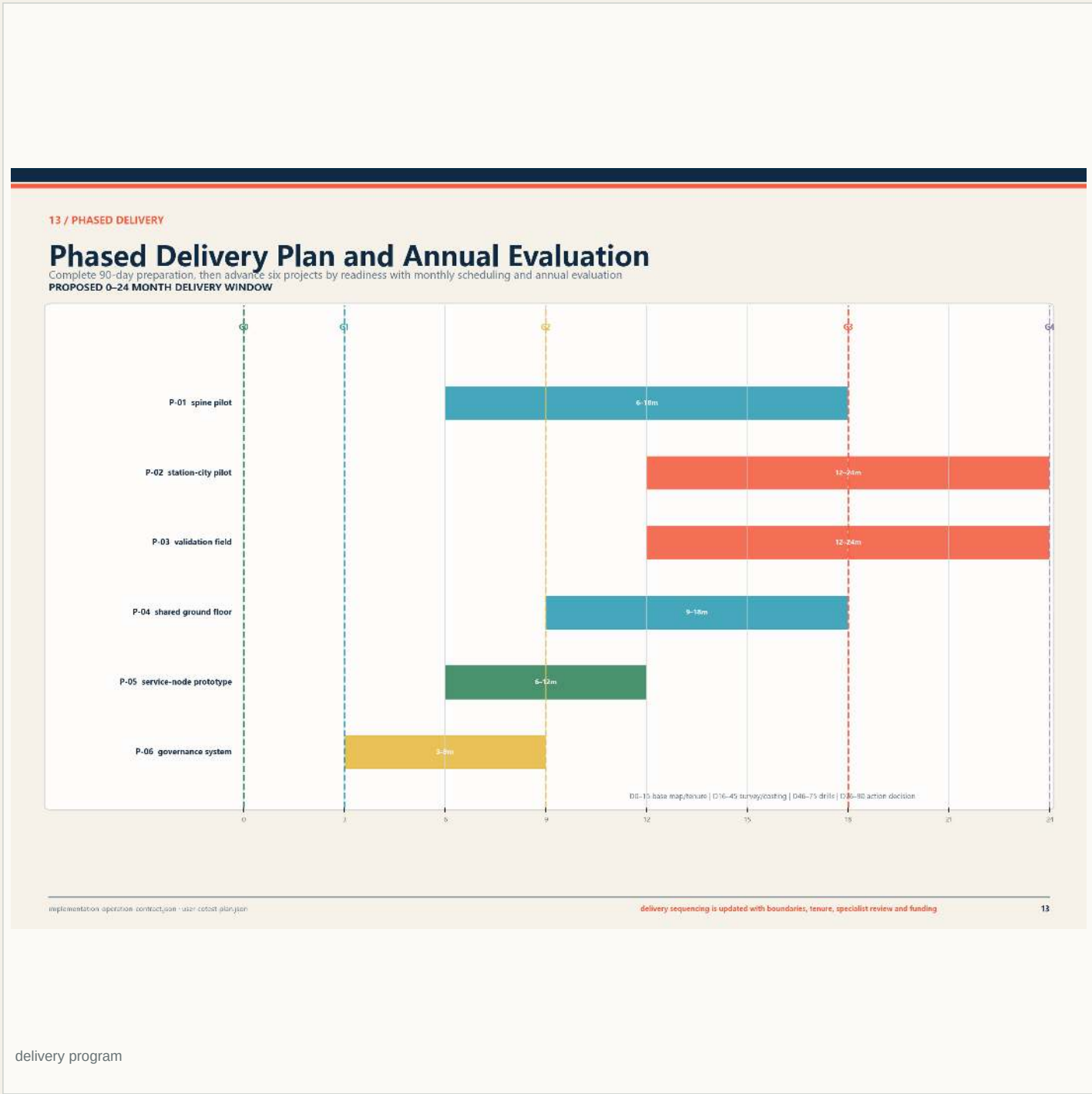
- coordinate rail protection and maintenance access
- integrate street limits, utilities and existing trees
- retain physical wayfinding and human service during outages

Controls still require coordination with formal road regulations, rail safety, fire access, utilities and existing trees

section dimensions are developed with rail safety, street limits, utilities and existing trees

0-24 Months: Near-term Packages and Delivery Sequence

Break design moves into reviewable, reversible and annually evaluated delivery packages



Inclusion and Annual Operations: AI Service Requires Human Fallback

The 8-80 equal-service contract spans facilities, information and operations

07 / INCLUSIVE SERVICES

Priority-user Needs and Project Rectification List

Complete priority-user walk-throughs before initiation and record issues, rectification duties and responses

REGISTERED USER ISSUES 0-3

Children and families	3		2		3		
Older people and caregivers	1	3	3				1
Disabled users	3		1		1	2	
Long-term residents			1		3		1
Business and small businesses	1		1	3	3		1
Students and young talent			3	3		1	
Commuters and visitors	3				1		
Facilities service and maintenance staff			1	2		1	3
Nearby campuses, parks and public institutions				1	2	1	
	mobility	rest + care	human service	learning + work	delivery burden	data + digital	maintenance + takeover

SURVEY + ACTION ARRANGEMENT

9类

priority-user groups

6项

participation steps

1套

rectification register

Coverage is calculated after population, employment and time-of-day flow baseline; current action to meet group issue lists.

inclusion ledger/guide - user cohort planning

all-age needs are integrated into consultation, design review and operating evaluation

07

inclusion and incidence

18 / ANNUAL OPERATION

Annual Events, Developer Community and Conversion

Four seasons, four working groups and a five-gate pathway share one annual programme

FOUR-SEASON ACTION CALENDAR

SPRING

URBAN PROBLEM SEASON

30 needs + 6 site walks

scenario register

SUMMER

OPEN VALIDATION SEASON

20 prototypes + 8 negative tests

test + connection records

AUTUMN

CIVIC INTELLIGENCE WEEK

12 co-tests + 4 workshops

co-test + open output/ dialogue

WINTER

ANNUAL GOVERNANCE ASSEMBLY

12 services + 6 packages reviewed

continue / correct / reduce / stop list

FOUR DEVELOPER WORKING GROUPS

01 public service + accessibility

02 model safety + evaluation

03 edge devices + robotics

04 urban data + maintenance

ANNUAL: 12 open workdays and 8 reusable outputs

FIVE-GATE CONVERSION PATHWAY

REGISTER

VALIDATE

CO-TEST

PROCURE

SCALE

NEXT GATE: rights clear, cost affordable, owner assigned and correction closed.

Events, site maintenance and staffed service use separate budgets.

annual operation programme/guide - growth handbook/guide

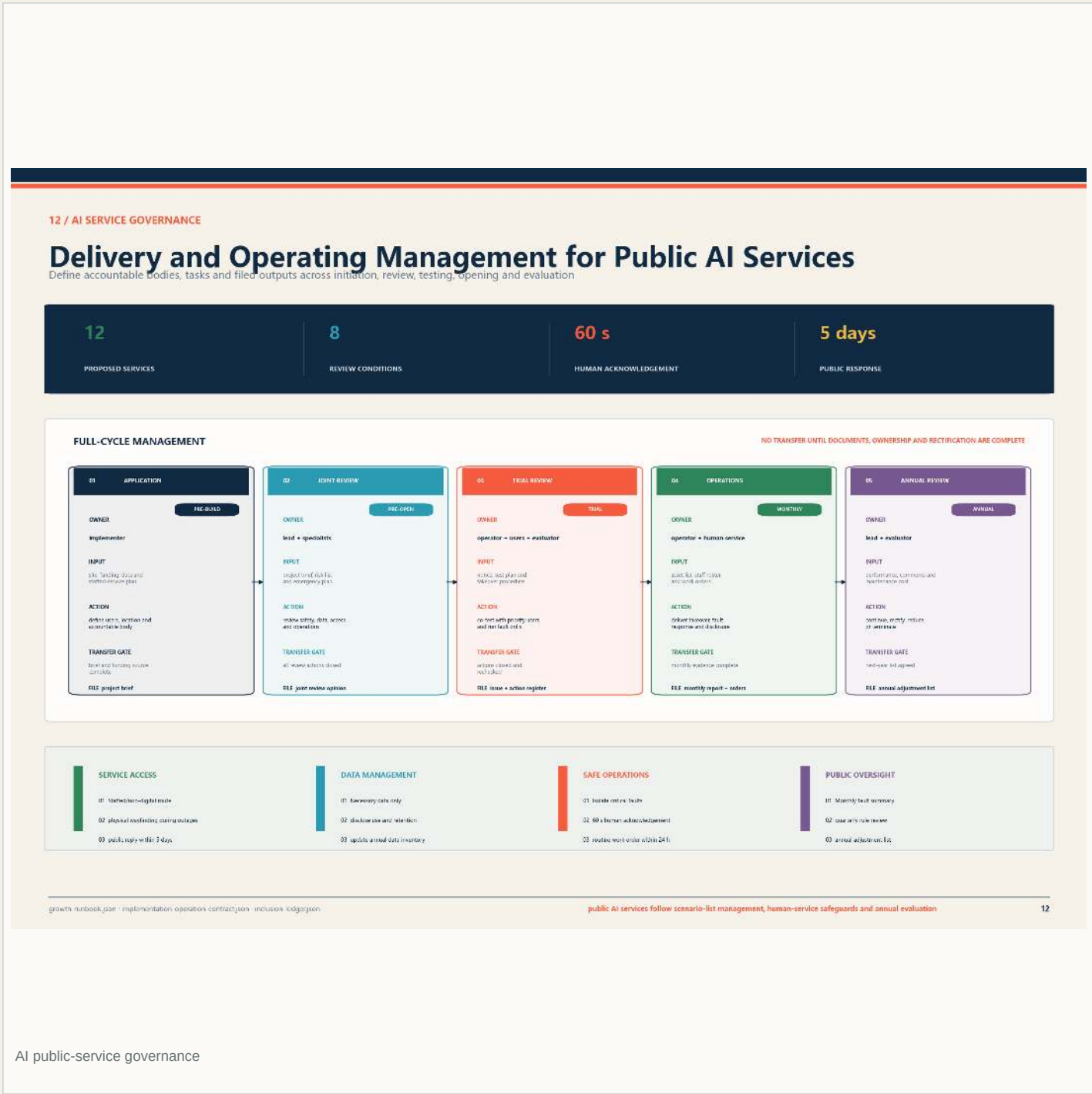
annual receipts are public; missed targets trigger correction, reduction or exit

18

annual operating cycle

AI Governance and Trial-operation Protocol

Technical launch, public feedback, incident tickets and annual review share one governance loop



Public Identity and Jingzhang Culture: Narrative Serves Everyday Space

Brand, cultural nodes and identity do not replace spatial evidence

14 / PUBLIC IDENTITY

Public-space Identity and Wayfinding Guidelines

A three-level system coordinates corridor gateways, direction signs and service confirmation



GROW WITH JINGZHANG

RAIL GATEWAY - WAYFINDING - HUMAN SERVICE - NIGHT IDENTITY

HIERARCHY + APPLICATION

A continuous, legible and all-age public information system

L1 | CORRIDOR GATEWAY

All street entrances and community entrances, cross identity and overall cohesion

L2 | DIRECTION SIGN

All route directions, stop destinations, distance, accessible route and interchange

L3 | SERVICE CONFIRMATION

All service nodes, transit, human desk, stop, and feedback route

INFORMATION + MAINTENANCE

- Separate project identity from government systems and maintain a clear information boundary
- Encourage service points, clear human service sign and non-digital access restrictions
- Local media duty, service status and stop information, easy and open support, emergency response
- Regular monthly media and public information work, and 24-hour

image2: public identity.jpg | brand system.json | implementation operation contract.json

public identity is designed, installed and maintained through the delivery program

14

identity system

17 / CULTURE + LANDMARKS

Cultural Narrative, Civic Landmarks and Recognition

Three narratives become everyday civic facilities, contributions receive annual review

THREE CIVIC NARRATIVES

INDEPENDENT ENGINEERING

1909—1949

Rail engineering, survey and construction archive

OPEN INNOVATION

1980—2025

Zhongguangyun research, open source and transfer

CIVIC INTELLIGENCE

2026—

Rights, duties and rules for AI public service

THREE CIVIC LANDMARKS

L-01

ORIGIN MILESTONE COURT

AI Origin Community

1909 / 1960 / 2026 archive and courses

1 installation / set + 1 course / year

L-02

OPEN MODEL GALLERY

Zhangshiyuan

Models, methods, failure cases and stop records

3D simulation records / year

L-03

CIVIC GOVERNANCE SIGNAL TOWER

Daxiongqi Public Hall

Live status for twelve public AI services

Live status + annual public report

OPEN CONTRIBUTION ARCHIVE

- problem discovery
- open tools + data
- safety + accessibility
- long-term maintenance

AWARD RULE

Public output, co-authorship and verifiable record; sponsorship alone does not qualify

culture: landmark system.json | brand system.json

landmarks combine access, rest, learning, wayfinding and staffed service

17

heritage / honour / pilgrimage nodes

Human-readable drawings; authoritative data remain in JSON/GeoJSON evidence files

AI Ecosystem and Regional Collaboration: External References for Programming

Cases and regional collaboration inform programming but are not existing-condition spatial data

16 / ECOSYSTEM + BENCHMARK

Benchmark Cases and AI Innovation Ecosystem

Six comparisons set policy choices; four stages and eight factors govern annual delivery

SIX BENCHMARKS: ADOPT / REJECT

Kendall Square

ADOPT: Continuous validation and transfer frontage

REJECT: Public service and R&D collaborations

Paris-Saclay

ADOPT: Synchronise transit and shared facilities

REJECT: Renewal and transacted planning

Kalasatama

ADOPT: Register tests, takeover and annual review

REJECT: Public-service continuity

Punggol Digital District

ADOPT: Share scenarios across park, university and community

REJECT: Translators and transferable interfaces

Barcelona Superblocks

ADOPT: Recover public space through mobility works

REJECT: Traffic assessment before capacity change

Vienna Gender Mainstreaming

ADOPT: Make priority group walks part of initiation

REJECT: Priority-group demand registers

FOUR-STAGE INNOVATION CHAIN

NEEDS

30 needs / year

VALIDATE

20 prototypes / year

CO-TEST

12 co-tests / year

CONVERT

8 conversion plans / year

COMMON RECEIPT: ID, provenance, safety review, correction and conversion result

EIGHT FACTOR RULES

LAND

Existing assets first

INDUSTRY

Five-sector service list

TALENT

Future-to-job pathway

DATA

Five data data register

SPACE

One-third paths to money conversion

FINANCE

Capital + 5-year OFEX

COMPUTE

Three-named maturing

SCENARIO

Open-call to walk update

ecosystem: policy join + six registered background cases

benchmarks inform method; annual targets enter programme control

16

ecosystem benchmark

11 / REGIONAL ACTION

Regional Annual Action Programme

Five collaboration units specify exchange, annual action, owner types and receipts

UNIT	ROLE	RESOURCE EXCHANGE	ANNUAL ACTION	RECEIPT
Beihai Community	neighbourhood co-test	public needs, research, devices, methods and service feedback	4 walks + 6 service improvements	issues + corrections
Future Science City	research scenario validation	public needs, research, devices, methods and service feedback	2 joint scenario challenges	challenge + test report
Huairou Science City	science-to-city transition	public needs, research, devices, methods and service feedback	2 course packs + 1 urban solution	course + feedback
Beijing Li-Town	robotics and manufacturing validation	public needs, research, devices, methods and service feedback	3 device tests + 1 supply-chain session	safety + conversion
Beijing-Tianjin-Hebei	interregional scenario network	public needs, research, devices, methods and service feedback	1 catalogue + 2 workshops + 3 off-site validations	catalogue + validation

DELIVERY RULE: enter the programme after annual agreement; register provenance, rights, budget, owner and exit.

regional action: program join + regional collaboration ledger join

collaboration is delivered through annual agreements and receipts

11

regional action network

Human-readable drawings; authoritative data remain in JSON/GeoJSON evidence files

Make Evidence Boundaries Part of the Scheme, Not a Footnote

Sentinel green and official-facility validation are complete; formal boundaries, tenure, utilities and field survey still require specialist confirmation

15 / BASIS + MATERIALS

Planning Basis and Material-use Statement

Materials are classified, sourced, managed by use and reviewed as an updatable planning record

8

MATERIAL CLASSES

30

PLANNING FIGURES

4

PDF DELIVERABLES

2

LANGUAGE SETS

MATERIAL	SOURCE / RIGHT	PLANNING USE	MANAGEMENT REQUIREMENT	STATUS
proposal text and translations	agreed authored for this submission	original submission text	policy and standard compliance (initial approval)	REQUESTED
technical figures and diagrams	generated locally from submission data by the package build script	agreed authored graphics	prominence, geometry is labelled and cannot be created as statutory mapping	TO PRODUCE
OpenStreetMap context	Downloaded by contributors	UKOS 3D ambience	background context only, no ownership or statutory relevance	USED
NASA POWER climate baseline	NASA POWER already API documented and reviewed online	public weather data with attribution	cross-check design against baseline, single writing design requires local data	PRODUCED
Blender corridor rendering	agreed generated from 3D modelled planning	agreed authored rendering	not an output or surveyed model	CONCEPT
generated concept scenes	Unreal Engine generated directly for this submission	submission concept imagery	design intent only, not identifiable real-world or official environment	TO BE REVIEWED
Three.js offline viewer	Three.js BGLS 1 loaded script	MIT license	license review, standard license package	USED
system fonts	Microsoft Windows or available GLE fonts	render only local font use	font files are not redistributed	PRODUCED

STATUTORY BASIS

Technical standards and best practices are included in the scheme register

TECHNICAL BASE

The working basis is updated with official boundaries, survey and amenity-related details

DELIVERABLE USE

Content is used to generate planning and design documents through specialist design outputs

sources:gear - rights:clearance:lodgement - reports:osyright_statement:mid

all materials follow source registration, use management and deliverable review

15

rights / source / clearance status

V6 / METRIC CONTRACT

Four Evidence Classes: Baseline, Design, Scenario, Delivery

Each number keeps its evidence type and verification condition; scenario effects are not built outcomes.

01 BASELINE / MODEL

192.6k

2025 WorldPop modelled population

17.9%

population with ≥ 1 service gap

26.23%

Sentinel-2 vegetation spectral proxy

02 DESIGN RESPONSE

12

service-node modules

320 m²

prototype area per node

1.0 km

CORRIDOR-04 first-phase window

03 SCENARIO EFFECT

39.4→61.9%

gap1 coverage: V1 → V2

51.5%

coverage of population with ≥ 2 gaps

99.86%

CORRIDOR-04 top-20 robustness

04 DELIVERY CONTROL

≥ 3 / ≥ 4 m

walk / two-way cycle clear width

≤ 150 m

effective rest-point spacing

≥ 70%

shade target at main stay spaces

100%

human service availability target

MEASURE → AUDIT → TRIGGER → ACCEPT

MEASURE

AUDIT

TRIGGER

ACCEPT

RE-CHECK TRIGGERS

formal redlines / tenure / utilities / rail safety / fire + accessibility / official facility data / field survey

All scenario effects are modelled/design-stage evidence, not built outcomes.

observed → response → impact → re-evaluation